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ORIGINAL ARTICLE

Image-based chip detection during turning

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Abstract

This study proposes a method to analyze chip formation using camera surveillance to enhance safety and efficiency in machine operations. The process involved the face and straight turning of a workpiece under the observation of a camera strategically placed within the workspace. The suggested algorithm carries out initial image preprocessing and edge detection, followed by background subtraction to isolate dynamic elements and filtering based on the size of the objects. Pre-determined masks are applied to eliminate overlaps with existing workspace objects, based on the tool's trajectory. The research validates that the applied technique effectively recognizes chips in both face and straight turning. Specific filtering techniques improve the algorithm's capability to detect even smaller chips, and it substantially reduces false alarms, laying the groundwork for long, continuous chip detection systems.

Keywords Lathe machining · Chip detection · Segmentation · Tool position · Operational safety

1 Introduction

With the advent of digital transformation in the industry, a wealth of image information is now available on the production process and the resulting product. Leveraging this data through image processing algorithms can lead to significant improvements in quality control and production efficiency, particularly through the use of closed-loop control systems.

In the case of cutting technology, the chip shape [1–3] and color [4, 5] still give us major information about the process. Monitoring this qualitative information during production enables automatic or semi-automatic intervention at the appropriate point, thereby increasing operational safety and production efficiency. Alarms and interventions based on chip-type monitoring are an important step toward the unattended production of CNC machines.

Video image analysis can enhance visual monitoring, making it more objective and automated. This improves accuracy and reduces errors that can arise from human factors such as mental fatigue or inappropriate professional acquaintances. Image processing-based methods offer the added advantage of expanding the capabilities of older systems with a one-time purchase.

When manufacturing with CNC cutting machines, industry players strive to keep maintenance costs low while maintaining the reliability of the production system at a high level [8]. In case CNC turning one of the critical points to avoid long continuous chip formation. Continuous chips can get stuck between the tool and the workpiece, mainly causing extra vibration and force, which can reduce workpiece's surface quality and accuracy. In case of a severe issue, it can damage the tool, the workpiece, and even the machine itself. Monitoring the chip formation, in case of unattended machining can increase the operational safety of production. Automatic intervention after chip shape segmentation can be an important milestone in the development of unattended production systems in the future [9, 10].

Using image processing methods, the shape of the cutting tool can be segmented, and its shape change can be detected [11–14]. The change in the tool shape can also be detected using indirect methods [15, 16], which can be further improved by using multi-sensory data fusion [10]. The output of the image processing algorithms related to the various production monitoring is an essential component of the digital twin models made on the CNC machine [17, 18]. With use of this technology, more and more accurate predictions can be achieved [19], the production process can be controlled, and the production parameters can be planned [20].

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Still, the chip formation analysis is one of the most common and informative methods for describing the cutting process performed on segmented chips [21]. The type of chip formed during machining depends on the cutting speed and feed value [22–24]. The chip shape (ribbon, tubular, spiral, conical, elemental, needle) can be predicted based on data of force sensor during the process [25]. Tool wear also affects changes in the formation and morphology of metal chips [16]. The chip formation is determined by the production parameters [26], the production parameters can be optimized. Simultaneously to ensure the appropriate chip formation. By properly controlling the machining parameters, a more even load on the workpiece can be achieved during production [27]. By correcting the value of the feed, the machine time can be reduced while providing suitable conditions, which also increases the efficiency of the production process. Adaptive control constraint (ACC) is an effective method of controlling the machining parameters to maintain the maximum working conditions [28]. In addition, NC program optimization based on the virtual model of the machining system is also an excellent method for increasing efficiency [29].

The aim of the research is monitoring chip formation with a camera and determine their character during machining process. The chips were formed on a CNC lathe. During the research, preprocessing edge smoothing was carried out. Then, during the chip detection, the location of the relevant area on the image was determined based on the displacement of the machining tool. The background and slowly moving parts were filtered out within these narrowed image regions using ghost images. Then, the remaining objects were further filtered based on geometric characteristics. As a result, the chips were successfully segmented.

Fig. 1 The camera was installed in the red-framed part of the workspace

Table 1 Parameters for test measurements

	Cutting depth [mm]	Feedrate [mm/revolution]
Test 1	0.0936	0.12445
Test 2	0.8327	0.12445
Test 3	1.5718	0.12445
Test 4	2.3109	0.12445
Test 5	3.05	0.12445
Test 6	1.5718	0.0839
Test 7	1.5718	0.104175
Test 8	1.5718	0.144725
Test 9	1.5718	0.165

2 Material and methods

Cutting tests were carried on machining Akira-Seiki SL25 C-axis lathe. Face turning and straight turning were performed on the workpiece, which was observed using a camera placed in a safe place in the workspace away from the workpiece (Fig. 1). The installed camera has not affected operational safety, so it can be a good solution for other CNC machines as well.

Legrand's CM7020 Dome Camera was used with a fixed camera angle. The other camera settings were also constant during machining. It is capable of making HD recordings at 30 fps. The camera is connected to a Hikvision NVR, where recordings made during measurements are stored. The recordings were evaluated after downloading them from the NVR via web browser module.

The test measurements were conductetd using an Akira-Seiki Performa SL25MC CNC C-axis turning machine—the test material comprised cold-drawn C15 steel bars with a diameter of 40 mm. For the test, a PCLNR2020K12 tool shank was fitted with a CNMG120408-MR7 TP100 SECO insert. The tests were made at constant cutting speed (165 m per minute). The cutting depth and the feedrate were changed, according to Table 1.

The method developed for processing camera images was implemented in the C++programming language using the OpenCV 4.5.5 function library. The algorithms run on a laptop (CPU: AMD Ryzen 7 5800H with Radeon Graphics 3.20 GHz, GPU: Nvidia GeForce RTX 3060).

3 Proposed algorithm

The algorithm that can do the chip segmentation becomes complex to reach better results. The main structure is shown in the Fig. 2.

In the case of a fixed-angle, constant-zoom camera region of interest (ROI) can be used. Region selection increases the performance since the algorithm only needs to run on one part of the entire image. 200 × 380 pixels image fragment was the ROI because of the position of the HD camera (Fig. 3).

Multiple Instance Learning (MIL) [30] was used to track the tool on consecutive images, which allowed the tool's current position to be accurately determined. The MIL tracker is a machine learning method applied in visual tracking that focuses on groups of instances instead of individual labeled data points. In tracking applications, this approach allows the

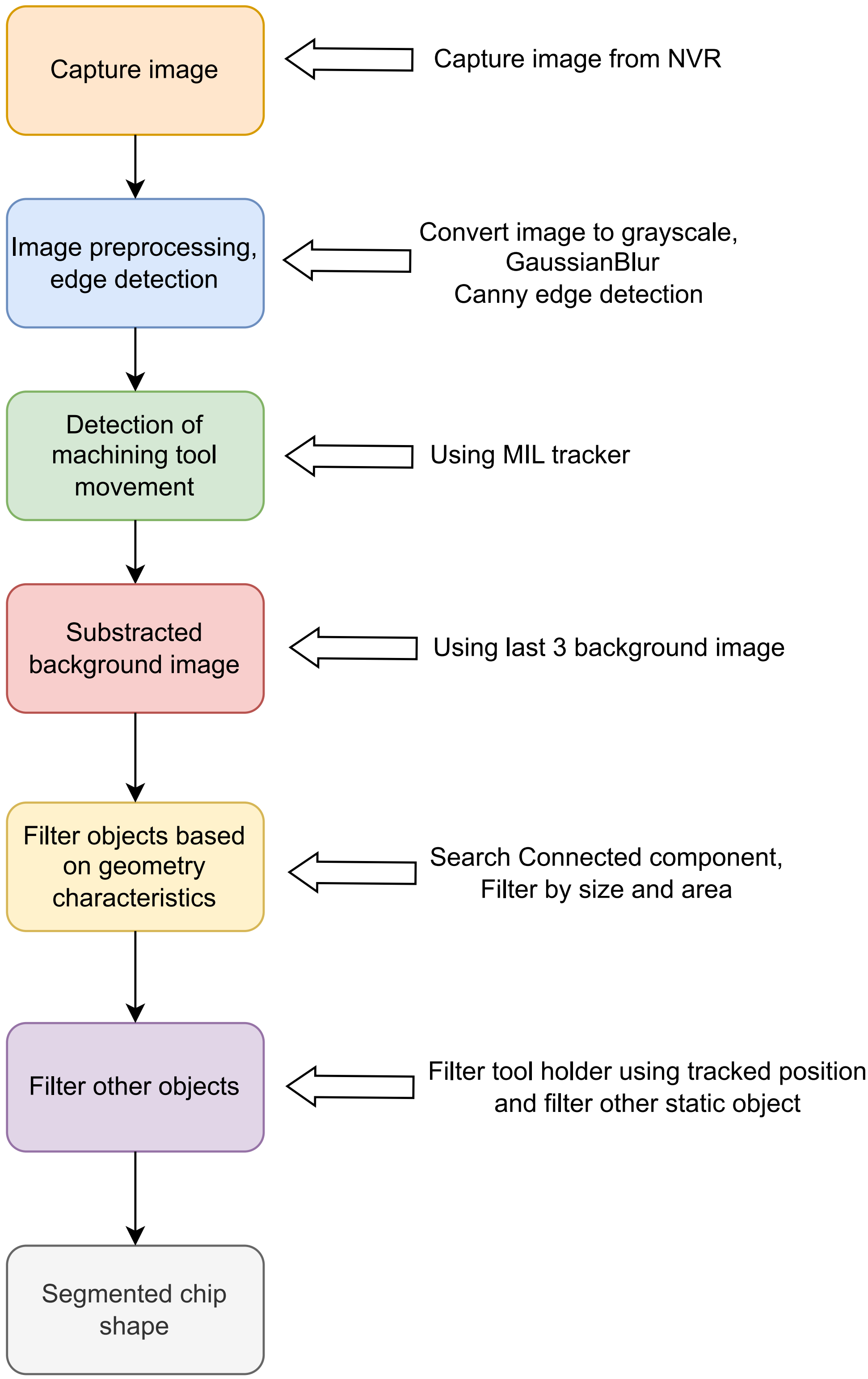

Fig. 2 The main steps of the proposed algorithm

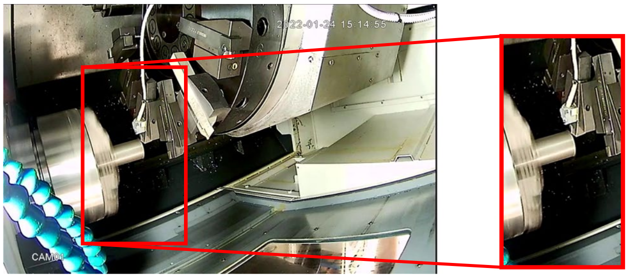

Fig. 3 The original frame (left) and the selected ROI with 200×380 pixels (right)

algorithm to handle uncertain or partially labeled data effectively by viewing each instance (such as an image patch) as belonging to either a positive or negative set. This is especially beneficial in situations where the precise position of the target object is unclear or obstructed. Recent developments in MIL tracking utilize deep learning strategies to improve robustness, flexibility, and precision in changing environments. In [30], the mathematical background of the MIL tracker can be found in detail.

Knowing the tool's current position allows the tool holder's edges to be filtered out of the recognized edges, and it also helps to reduce the uncertainty of movement tracking. To filter out the tool holder edges, two previously sampled masks are used at a distance of 5 pixels along the axis of both cords from the tool position, and the closest mask is found for a given tool position. The permanent, static components of the image can be filtered by subtracting the ghost background image from the current image. These components are, for example, the martial clamps or the hanging pipes. The remaining edges can be further filtered based on area. As the last step of the algorithm, the obtained mask is applied to the original input image. Figure 4 shows the step of the image processing; the difference between the two steps indicates a color, which represents the function according to Fig. 2.

4 Results

4.1 Tool positioning

The purpose of the applied ROI is to quickly determine the position of the tool and the insert where the chip is formed. The current position of the tool is relevant if it is close enough to the workpiece to perform work. The tool position, if it is far enough from the workpiece, no machining takes place. For further analysis, it is sufficient to store the chip shape only during machining, so a significant saving of storage space can be achieved, and a conclusion can be drawn when the machine time is not being used.

The main events in the test video were:

- 0–116 frame ID (I.): The tool is moving in a radial direction (face turning), chipping is taking place.
- 116–173 frame ID (II): The tool moves away from the workpiece in the axial direction and then in the radial direction, no machining is performed.
- 173–323 frame ID (III): Tool moves in axial direction (straight turning – approaching chuck), machining occurs.
- 324–367 frame ID (IV): Tool moves away from the workpiece in radial direction, then moves to starting position in axial direction, no machining.
- 367–467 frame ID (V): Tool moves in axial direction (straight turning), machining is performed.

The MIL tracker was used to track the tool. The position is calculated by the tracker on each frame. As can be seen in Fig. 5, the ranges determined from the video (human recognition – previous list of frames) and the tool path show good agreement. Chipping is done on the parts marked the blue rectangles on the figure.

4.2 Different chip shapes

Based on the test measurements, the proposed method has segmented the chips well and visually shows an excellent match with the expected one (Fig. 6).

It can be seen that the segmentation works well when the chips form long continuous chips (Fig. 6a, d) and also when short chips are formed (Fig. 6b, d). The chip tracking also provides good detection when the chip is already further away from the cutting zone (Fig. 6e, f), i.e., the chip is far from the insert. The measurement shows that the proposed method segments well the different chip types and shapes too (Fig. 6a, b, c). The chip segmentation is also effective when the chips move in different directions (Fig. 6d, e, f).

4.3 Overlapping

The method does not properly segment some small, insufficiently contrasting chips. However, this is not disturbing as the result will be used for quantitative purposes in the future. When determining the length of a chip, it is better not to detect all chips while at the same time avoiding misidentifying them as other chips. When a running grip is generated, it tends to become larger during machining, which the algorithm detects well.

If edges overlap with other objects, then the chip is not segmented, because the method minimizes false detections. Some examples are shown in Fig. 6: chip on the workpiece (b), completely overlapping chip with the tool holder (a),

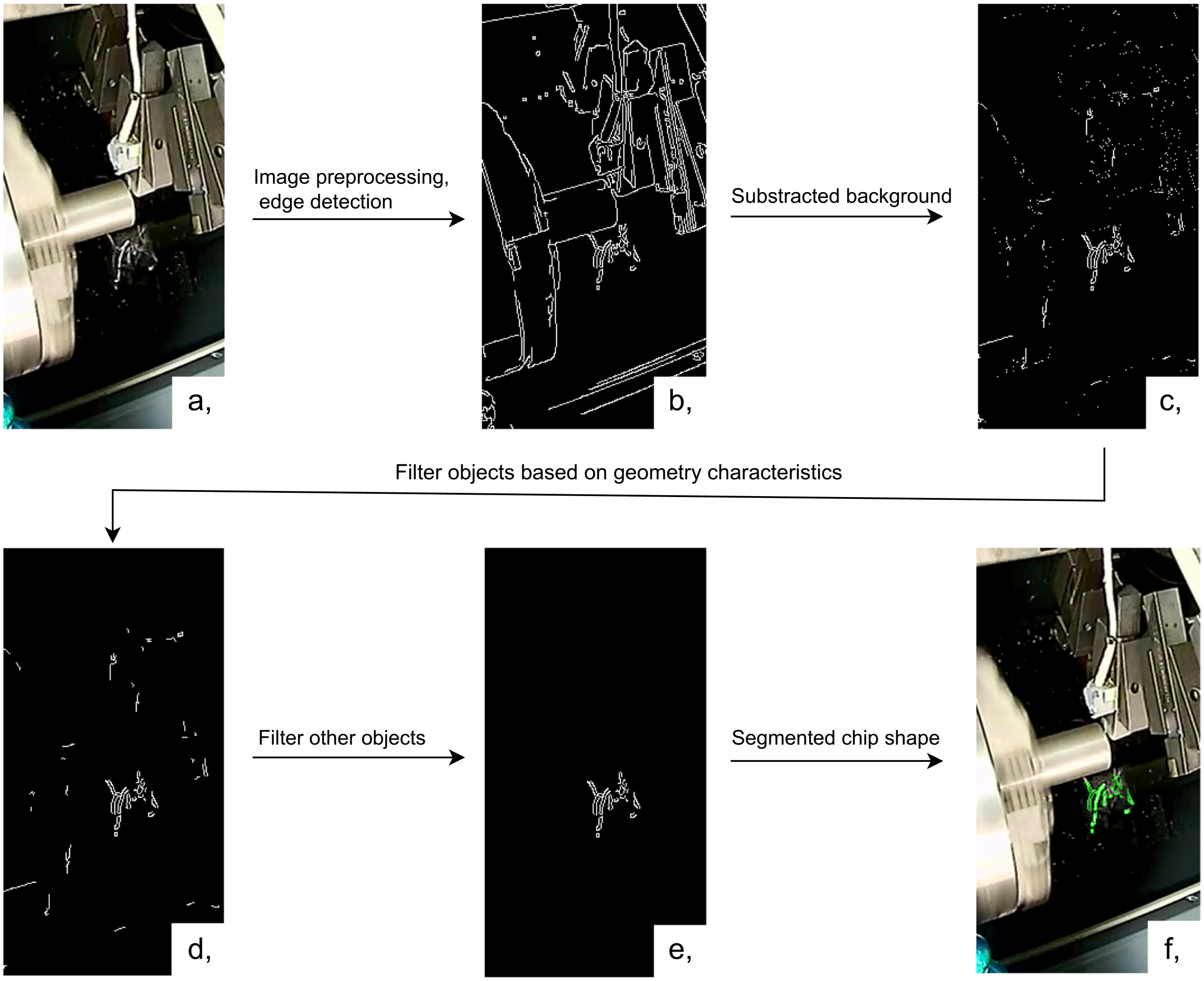

Fig. 4 Presentation of the effects of individual steps of the algorithm in an example image (**a**), original image (**b**), image preprocessing, edge detection **c**, subtracted background, **d**, filter other objects e, segmented chip shape

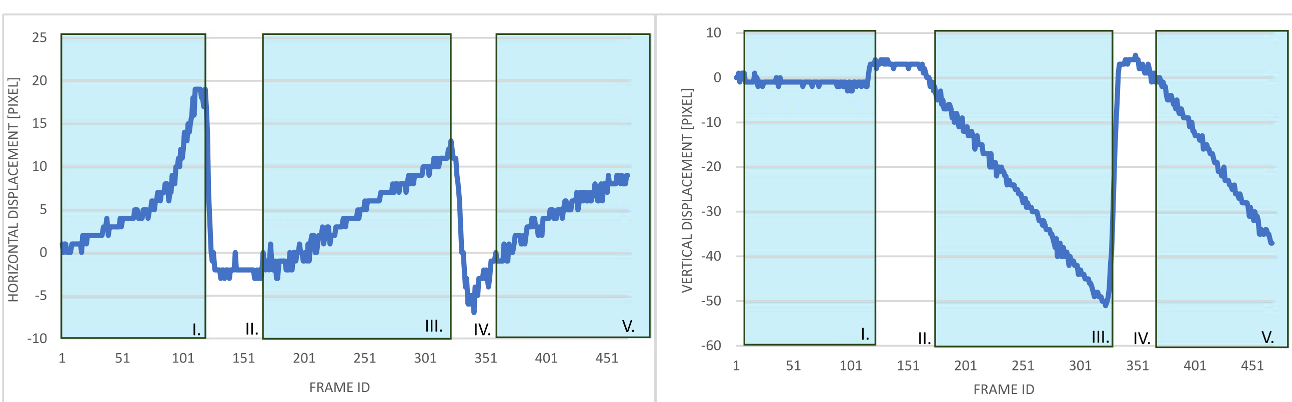

Fig. 5 The horizontal and vertical displacement of the tool at each frame

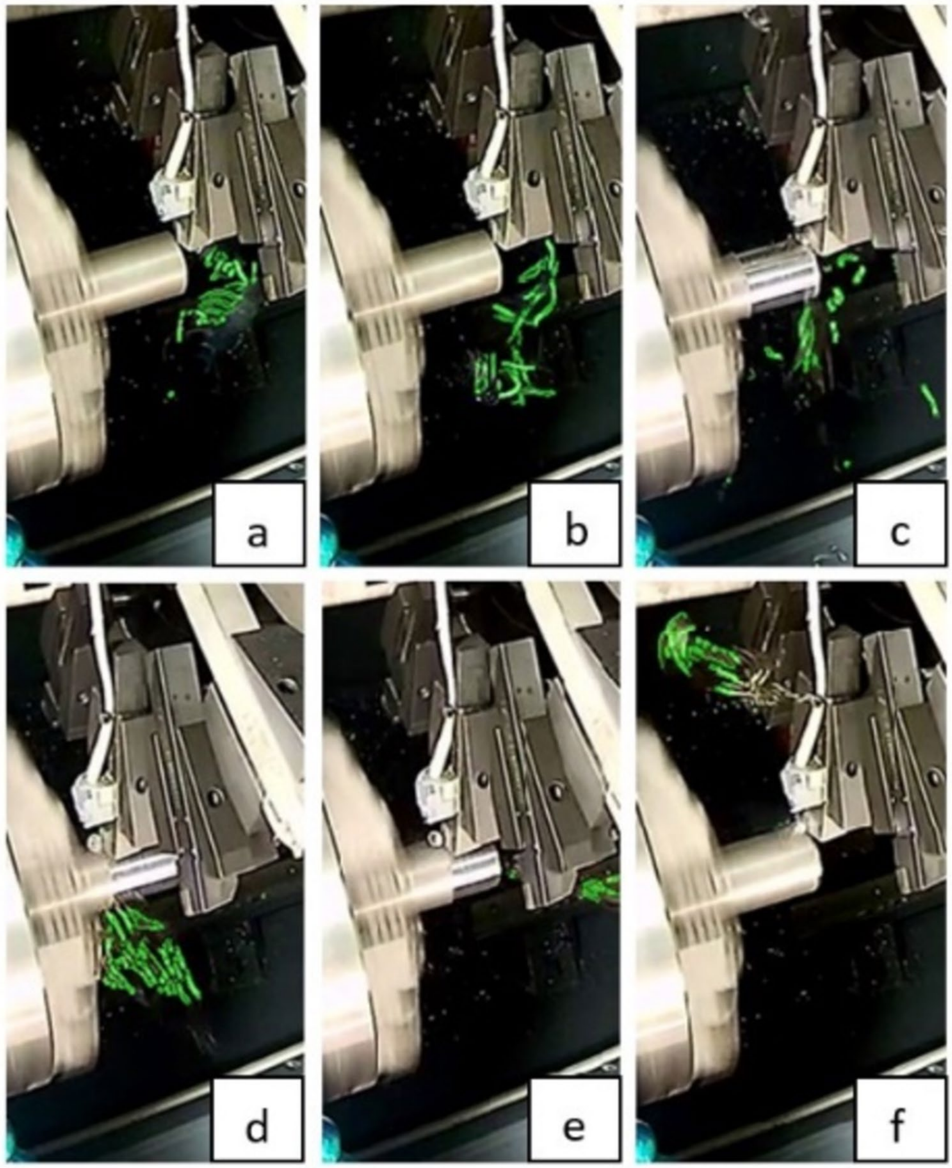

Fig. 6 Presentation of some special cases on some selected frames. The segmented chips are marked in green in the figure. (**a**–**d**) chips of different sizes, (**e**–**f**) chips moving forward and backward

partially overlapping chip (c). When we did not remove overlapping objects, some parts from other objects were still included despite the background subtraction. The method did not consider chips where the chips had become detached within the workpiece.

4.4 Chip segmentation

Tool clamping and workpiece edges were well filtered by background correction. However, in some cases (e.g., very fast movement, glancing) it did not work properly, giving false positives (Fig. 6d). By excluding the image area of objects, detection efficiency was reduced (Fig. 6f), with a relatively significantly better number of badly detected chips. The next step in the research will be the classification of chip shapes. The key parameter is to detect the long continuous chip. So, it was important to minimize the number of false-positive image patches in the training base. In addition, it is also better from an operational safety point of view to have fewer false alarms.

The effect of background subtraction increases the number of segmented objects. The explanation for this is that due to the movement, the edges do not overlap perfectly due to the movement of the tool, but instead fall apart into several smaller pieces (Fig. 4c). Optimally, this can be filtered based on geometry features in general. However, in some cases, more than area-based filtering may be required because new edges are introduced from the tool clamp or the piece of material to be machined (Fig. 4d). This problem was addressed by excluding static stationary parts and image regions associated with tool position-based tool clamps, eliminating false segmentation (Fig. 4e, f). The filter of the objects from the image significantly reduced the number of objects (Fig. 7).

In the edge detection performed on the frames, the number of edges in the machining and non-machining situations falls roughly in the same range (Fig. 8a). After the background correction, the number of edges increases in the case of machining sections (Fig. 8b). This happens for two reasons: on the one hand, due to the segmented chip edges, and on the other hand, if the previous background fits worse, the number of edges generated from the tool holder will increase. After the correction, there are fewer edges in the unprocessed case than in the initial case, so the filtering of the objects is successful.

Applying area-based filtering on the objects, the number of detected objects (Fig. 8c) is reduced, but the curve does not change significantly. This step removes small objects created during background subtraction, which are fitted imperfectly. The number of detected objects can be further reduced by mask-based filtering of objects and repeated area-based filtering (Fig. 8d). It can be seen in the figure

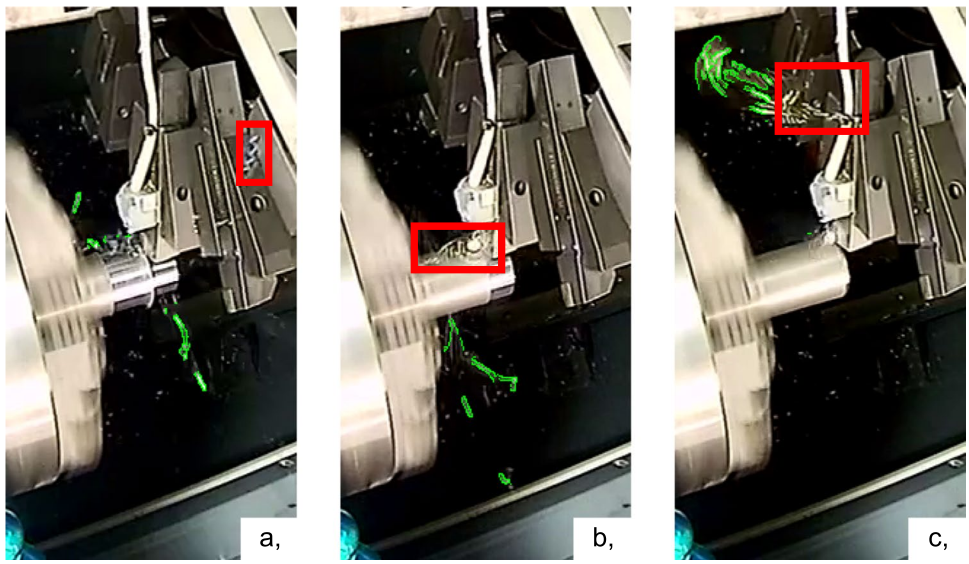

Fig. 7 Some examples of overlapping objects

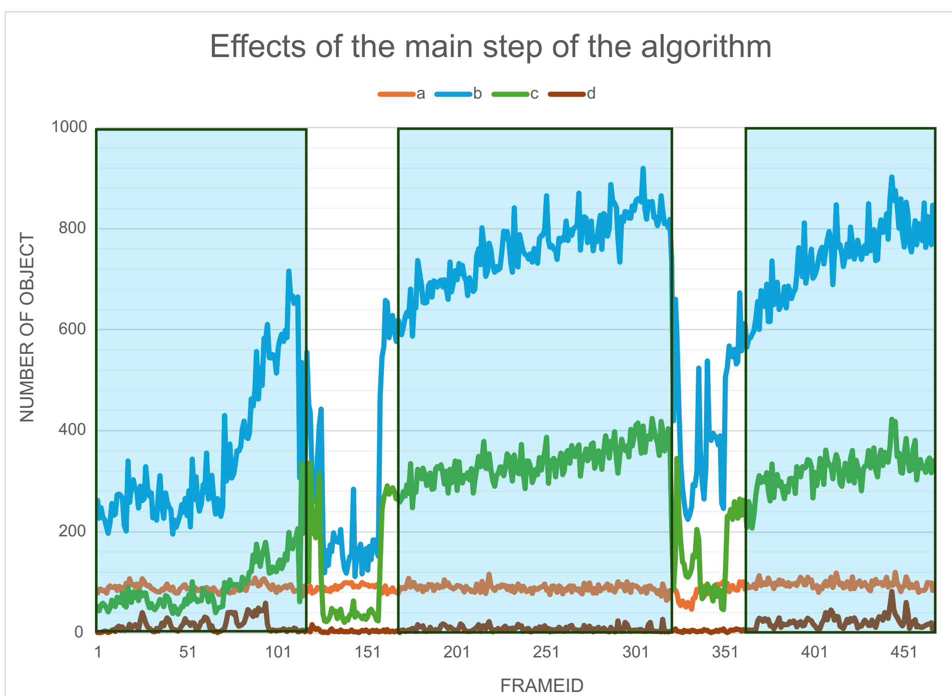

Fig. 8 The main step of algorithm: (**a**) preprocessing, (**b**) subtracting background, (**c**) area based filtering, (**d**) filtering other objects

that the number of detected objects decreased to almost 0 in the non-machining section. Due to technical reasons (different chip formation), the number of objects detected in the three machining ranges is different. With this final step, the algorithm can work well and realistically.

5 Summary

The basic algorithms (e.g., tracker, edge detection, connected component) are used unchanged from the OpenCV function library. The currently used methods deal with stationary chips, they do not deal with the analysis of moving chips, so the applied approach is promising. Our method is unique, as it uses individual building blocks one after the other in the right order for the solution of chip segmentation. This unique approach ensures the effectiveness and novelty of our method. When analyzing the camera images recording the machining process, the chip shapes can be safely segmented, on which many industrial applications can be built later. The integration of the camera does not affect the operation of the CNC machine, as it functions independently. Additionally, this solution can be implemented cost-effectively within existing systems, offering a quick return on investment.

The results achieved with the MIL tracker hold the potential to improve production processes, making them safer and more efficient. By anticipating the need for tool replacement based on the characteristics of the chips, which change due to tool wear, the MIL tracker enables timely interventions to prevent material loss and machine downtime caused by tool breakage. The ability to display segmented chips on an external LCD or record them in a database for expert analysis, and the prospect of automatic interventions based on this data, further underscore the practical applications of the proposed method. With further enhancements, the proposed method could even support increased production efficiency by potentially replacing or assisting the workforce.

By using the proposed method, reliable chip detection is possible, which is a result that can be easily used since services based on fake detection can cause unnecessary production stoppages. Based on the experimental results, the method can be used for both face turning and straight turning on the workpiece. Unfortunately, it is not possible to use a simple algorithm for chip detection as there are many challenges to overcome. In the workspace observed with a camera fixed at a fixed angle, fast movements and changing light conditions affect edge detection. When choosing the applied filters, a specific characteristic of the cutting process is taken into account, for example, chips can be generated when the tool is close to the workpiece. In this case, the research aims to detect long-flowing chips for later analysis. Therefore, it is acceptable to discard some small chips and overlap chips with other objects for better usability and greater safety.

The algorithm can be utilized for minimal lubrication systems as well as those without cooling. When using an emulsion, detecting chips in the area of the tool during recording may not be feasible; however, it is anticipated that chips flying outside this zone can still be detected and analyzed. The method will be ineffective if water vapor condenses on the camera during processing.

In future research, we plan to explore the extension of this method to various types of CNC systems in greater detail. In addition, based on the characterization of the moving chips, we would to create closed loop control system of the CNC machine. In order to perform the appropriate interventions automatically to setting the optimal machining parameters (i.e., feedrate, cutting speed). The current results were achieved with a low-cost camera. In the future, it would be worthwhile to examine the results achieved with a better image quality camera.

Author contribution The method proposed in the research was designed, implemented, and tested by Tamas Filep. During the evaluation of the partial results and the final results, Mátyás Andó and Béla J. gave useful advice and helped with their ideas. Mátyás Andó provided the technical background knowledge for the research. In the IT section, Béla J. Szekeres's insights were particularly useful. The manuscript of the article was written by Tamas Filep in consultation with Mátyás Andó and Béla J. Szekeres. The review and final tuning of the article was carried out by Mátyás Andó and Béla J. Szekeres. All authors discussed the results and contributed to the final manuscript.

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Declarations

Competing interests The authors declare no competing interests.

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